[NERC 2026] Learning to Accelerate Task and Motion Planning for Forceful Human Limb Manipulation

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Robotic assistance in domains such as caregiving and rehabilitation requires physical human-robot interaction (pHRI), often involving direct, forceful contact between the robot and the human body. Providing such assistance to an unseen person over a long horizon cannot be handled by a low-level controller or a single learned policy alone: the robot must also plan where to make contact and which sub-tasks to perform, subject to that person's physical constraints. Forceful manipulation has been formulated as Task and Motion Planning (TAMP) [1] or as a Constraint Satisfaction Problem (CSP) [2], but has primarily been used for object manipulation such as opening childproof bottles or slicing food. Planning work involving humans typically has the person exert force through a tool while the robot holds the object [3], rather than the robot contacting the human body directly.

We formulate forceful human limb manipulation as a TAMP problem. Once the robot is in contact, it must satisfy kinematic and dynamic constraints on both bodies: joint and actuator torque limits and collision avoidance for the robot, and, on the human side, the range of motion (RoM) of each joint and the torque within that person's tolerable limits. Unlike the robot's limits, these human limits differ across people and shift over time for the same person, so they cannot be hard-coded and must instead be learned from the user. State-of-the-art sampling-based TAMP planners [4] struggle to find configurations and torque trajectories satisfying such constraints, since they must sample without knowing the person: unknown RoM and unknown muscle tone yield infeasible and unsafe motions. This raises our research question: given a new person, how can a robot learn that person's physical parameters and adapt its planner to them quickly?

To overcome this issue, we propose a framework in which the robot repeatedly learns about a new person and learns to plan for them. Since the person's physical parameters, such as the RoM of each joint, are partially observable, the robot maintains a belief over them. The framework consists of four steps:

- **Information gathering:** The robot asks the person to reach a pose near a believed RoM boundary and collects the human's response.

- **Human model learning:** The robot updates its belief over the person's parameters based on whether the person succeeds.
- **Learning to plan:** The robot improves its planner offline: during a "sleep" phase, it solves sampled tasks for this person with a slow planner and uses those solutions to refine the planner's learnable parameters. In this task, the learnable parameter is the initial torque distribution supplied to an MPC layer that enforces the joint and torque limits of both bodies.
- **Planning and execution:** The robot plans and executes new tasks for that person with the personalized planner.

We evaluate on diverse limb manipulation task variants in simulation, in which a mobile manipulator moves a person's limb to a goal configuration across settings such as the person lying in bed or seated in a wheelchair. Each variant is evaluated across a population of simulated humans with differing physical parameters. Compared to a baseline sampling-based TAMP planner [4], our learned planner reduces planning time without violating each person's individual RoM limits.

References

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